

VERMILION REGIONAL, IL, USA

WMO: 722076

Lat: 40.200N

Lon: 87.600W

Elev: 202

StdP: 98.92

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 94891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-15.0	-22.1	0.5	-17.0	-19.4	0.7	-14.4	12.6	0.6	11.6	1.2	2.9	300	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.3	23.3	32.0	23.5	30.5	22.9	26.2	30.8	25.2	29.6	24.3	28.4	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.5	29.2	23.9	19.2	27.8	22.9	18.1	26.8	82.7	30.9	78.0	29.8	74.4	28.4	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.5	-22.6	34.6	3.2	1.8	-25.0	35.9	-26.8	37.0	-28.7	38.0	-31.0	39.4		
(5)			-22.4	28.1	3.6	0.7	-25.0	28.6	-27.1	29.0	-29.1	29.4	-31.7	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-3.0	-1.7	5.1	11.8	18.1	22.5	23.4	22.3	19.6	12.8	5.7	0.0		
	DBStd	10.83	6.99	6.70	6.68	5.35	5.01	3.24	3.02	3.08	4.13	5.34	5.57	5.82		
	HDD10.0	1455	405	329	184	42	3	0	0	0	0	30	150	311		
	HDD18.3	3086	662	560	413	205	69	6	2	4	33	186	378	567		
	CDD10.0	1989	2	3	32	95	253	374	417	382	289	118	22	2		
	CDD18.3	579	0	0	3	8	62	129	161	128	72	16	1	0		
	CDH23.3	5372	0	0	25	97	565	1134	1527	1167	709	147	1	0		
	CDH26.7	1809	0	0	3	14	177	378	550	390	257	40	0	0		
(14) Wind	WSAvg	3.7	4.7	4.4	4.5	4.7	3.9	3.2	2.3	2.0	2.4	3.5	4.2	4.2		
(15) Precipitation	PrecAvg	989	53	47	76	98	107	107	111	93	74	77	81	66		
	PrecMax	1378	186	160	162	276	253	316	325	229	220	218	260	178		
	PrecMin	610	2	10	20	14	4	8	16	30	1	8	13	6		
	PrecStd	166	38	30	38	55	53	62	68	49	47	50	49	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	17.7	26.8	28.0	32.6	33.5	36.0	34.4	34.2	31.0	22.3	16.7		
		MCWB	14.8	13.2	18.4	16.0	18.3	22.9	24.6	25.2	21.8	22.3	14.3	14.5		
	2%	DB	12.8	14.1	22.6	26.2	30.5	32.2	32.7	32.4	32.2	27.3	19.1	13.0		
		MCWB	10.9	12.0	16.5	16.0	20.2	22.9	24.9	23.9	23.1	18.6	14.4	11.2		
	5%	DB	9.9	11.1	18.6	23.8	28.7	30.7	31.3	30.4	29.8	24.2	16.5	10.9		
		MCWB	8.0	8.8	14.0	15.4	19.2	22.8	24.7	23.5	21.5	17.4	13.1	9.5		
	10%	DB	6.3	7.7	15.7	21.3	26.6	28.9	29.8	28.8	27.4	22.0	14.0	7.9		
		MCWB	4.5	5.6	11.7	14.6	18.6	21.5	23.8	22.9	19.8	16.8	10.5	6.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	14.9	18.9	19.6	23.4	26.3	27.8	27.1	25.1	23.3	17.8	15.0		
		MCDB	16.7	16.5	25.0	25.0	28.7	30.8	31.6	32.6	31.1	29.2	20.2	16.4		
	2%	WB	11.3	11.8	17.0	18.0	22.2	25.0	26.6	25.7	23.9	21.2	15.4	11.7		
		MCDB	12.5	13.7	22.0	22.9	27.2	29.8	31.0	30.4	29.4	25.0	18.0	12.8		
	5%	WB	8.2	8.8	15.0	16.7	21.2	24.0	25.6	24.7	22.8	19.4	13.3	9.1		
		MCDB	9.4	10.6	18.2	21.5	26.0	28.4	29.7	28.9	27.3	22.4	16.3	10.2		
	10%	WB	4.8	5.6	12.3	15.4	20.2	23.0	24.6	23.8	21.7	17.2	11.1	6.5		
		MCDB	6.5	7.8	15.1	20.5	24.8	27.1	28.4	27.5	25.6	21.0	13.6	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	9.9	11.7	11.3	10.8	10.9	11.7	12.8	11.6	9.2	7.4		
		MCDBR	11.1	12.3	12.9	14.7	13.8	12.2	12.2	13.1	15.0	14.7	12.4	10.6		
	5% WB	MCWBR	10.0	10.2	8.0	7.6	6.0	5.7	6.0	6.8	6.5	7.5	8.3	9.4		
		MCWBR	10.7	11.9	11.8	12.2	11.3	10.2	10.5	10.9	12.1	12.0	11.1	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.317	0.353	0.398	0.435	0.439	0.456	0.458	0.421	0.365	0.319	0.299		
	taud		2.433	2.377	2.360	2.241	2.192	2.220	2.162	2.149	2.230	2.381	2.468	2.456		
	Ebn at Noon		865	898	899	875	847	842	822	809	815	830	836	836		
	Edh at Noon		80	98	111	134	144	140	148	146	126	97	77	73		
(44) All-Sky Solar Radiation	RadAvg		1.76	2.65	3.65	4.73	5.40	6.20	6.11	5.47	4.58	3.14	2.09	1.48		
	RadStd		0.19	0.27	0.28	0.35	0.47	0.37	0.33	0.32	0.34	0.31	0.20	0.17		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page